MILITARY SPECIFICATION

TRAILER, BOMB, MHU-123/M

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes the requirements for one type of bomb trailer, designated as MHU-123/M, hereafter referred to as the trailer.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-S-681	Steel Castings

Military

MIL-D-1000	Drawings, Engineering And Associated Lists
MIL-C-3774	Crates, Wood; Open 12,000- and 16,000-Pound Capacity
MIL-T-5021	Tests; Aircraft And Missile Welding Operator's Qualification
MIL-H-5606	Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance
MIL-L-6086	Lubricating Oil, Gear, Petroleum Base
MIL-E-6843	Electrodes, Welding, Covered, Low-Alloy Steel (Primarily For Aircraft And Weapon Applications)
MIL-W-6858	Welding, Resistance: Aluminum, Magnesium, Nonhardening Steels Or Alloys, Nickel Alloys, Heat-Resisting Alloys, And Titanium Alloys: Spot And Seam
MIL-I-6865	Inspection, Radiographic
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements For
MIL-A-8421	Air Transportability Requirements, General Specification For
MIL-E-8697	Electrodes, Welding, Covered, Low-Hydrogen, Heat-Treatable Steel
MIL-G-21164	Grease, Molybdenum DiSulfide, For Low And High Temperature

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STANDARDS

Military

MIL-STD-22	Welded Joint Design
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-162	Materials Handling Equipment: Preparation For Shipment, Storage, Cyclic Maintenance, Routine Testing And Processing
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation Of

Industry

AWS Standard AWS A2.0-58 Welding Symbols

DRAWINGS

Air Force

"J" 7037580	Trailer, Life, Bomb, MHU-7/M to MHU-123/M, Modification of

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society For Testing And Materials

ASTM - Radiographic Standards for Steel Castings (X-Rays)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Design and construction. The trailer design and construction shall conform to USAF Drawing 7037580 and subsidiary drawings.

3.2.1 Weight. The weight of the trailer shall not exceed 10,500 pounds.

3.2.2 Dimensions. The trailer shall be adjustable in width. The width in narrow position shall not exceed 87 inches. The width in extended position shall not be less than 124 inches. The length, without towbar attached, shall not exceed 216 inches.

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3.2.3 Commercial parts. Commercial parts having properties which equal or exceed standard parts may be used where, on the date of invitation for bids, there are no standard parts available. In any case, commercial utility parts, like screws, bolts, nuts, cotter pins, having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.2.4 Standard parts. With the exception in 3.2.3, MS and AN standard parts shall be used. They shall be identified on the drawings by their part numbers.

3.2.5 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-1000 shall govern changes in the manufacturer's part numbers.

3.2.6 Welding. All welding shall be in accordance with MIL-W-6858, AWS A2.0-58, and MIL-STD-22. Unless otherwise indicated on detail drawings, the welding rod shall be in accordance with MIL-E-6843, Class A for sheet metal parts and MIL-E-8697, Class E-10016 for all parts other than sheet metal. All welding shall be performed by personnel certified in accordance with MIL-T-5021.

3.2.7 Power requirements. The drive motor for the trailer hydraulic system shall have a nominal rating of 5.00 horsepower (hp) and shall operate from a power source of 208 volts, 400 cycle, 3 phase. To supply this power, any generator of required characteristics and output of at least 7.5 kilovoltampere (KVA) may be used.

3.2.8 Steel castings. Unless otherwise indicated on detail drawings, all steel castings shall be in accordance with QQ-S-681.

3.2.8.1 Steel casting machined surface. All parts having a machined finish symbol callout will be free of flaws in such surface. Flaws include such defects as cracks, blowholes, checks, ridges, and scratches.

3.3 Performance requirements.

3.3.1 Reliability. The MHU-123/M trailer shall have a Mean Time to Failure (MTTF) of 500 miles and 5000 complete lift cycles with a minimum reliability of 90 percent for a mission duration of 20 miles and 10 lift cycles at a 90 percent confidence level.

3.3.2 Continuity. The power and lighting circuits shall satisfy the continuity test described in 4.6.

3.3.3 Hydraulic lift system. The hydraulic lift system, using hydraulic fluid specified by MIL-H-5606, shall be capable of lifting and holding a load weighing 27,500 pounds to a position where the bottom of the lifting beams will be not less than 84 inches above the ground. Lifting time shall not exceed 180 seconds. Lowering time of the lifting arms, unloaded, from the fully extended to the fully retracted position shall not exceed 180 seconds. The hydraulic system shall function in such a manner that height differential between the lifting beams shall not exceed 1-3/8 inches. The auxiliary hand pumps shall

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be capable of raising a 27,500 pound store to the maximum height. The controls of the lift system shall be such that either set of lifting arms can be controlled separately and complete control can be maintained throughout entire lifting or lowering operation with rated load imposed. It shall be possible to raise or lower the loaded lifting beams in increments of 1/8 inches.

3.3.4 Load attitude adjustments. The trailer shall be capable of adjusting the attitude of the rated load as follows:

a. Ten degrees clockwise and counter-clockwise about the longitudinal axis of the load (roll axis).

b. Tilt the nose of the load 5 degrees up and 3 degrees down.

c. To move the load 5 degrees right and 5 degrees left about its own vertical axis (yaw axis).

d. To move the load 5 inches forward and 5 inches rearward from center of the trailer.

3.3.5 Brakes.

3.3.5.1 Parking brakes. The parking brakes shall be capable of stopping and holding the trailer, loaded with a 27,500 pound test load, on a 13 degree slope. These brakes shall be easily operated by personnel walking or standing alongside the trailer during the aircraft loading operations.

3.3.5.2 Service brakes. The pneumatic-hydraulic service brakes shall have the following capabilities:

a. To stop the trailer, with rated load, within 50 feet when traveling at a speed of 20 miles per hour on a dry, level concrete surface.

b. To perform the above stopping action repeatedly within a short period of time without damage or distortion.

c. To retard the speed of the trailer, with rated load to 5 miles per hour for a distance of one-half mile while traveling down a 10 percent slope without damage, distortion, or diminishing effectiveness of braking action.

3.3.5.3 Break-away brakes. The trailer shall be provided with break-away brakes that will automatically stop the fully loaded trailer within the limits described in service brake requirements in the event of breaking loose from the prime mover during towing operations.

3.3.6 Steering. The front wheels of the fully loaded trailer, when being towed shall be capable of turning through an angle of 45 degrees right and left of center.

3.3.7 Castering. When detached from the towing vehicle, the towbar removed, and the weight of the trailer removed from the wheels, the wheels shall be capable of being manually castered 90 degrees with respect to the right of the center line of the trailer.

3.3.8 Mobility. The trailer, carrying loads up to 27,500 pounds, shall meet the type II mobility requirements of MIL-M-8090.

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3.3.9 Proof load. Trailers subjected to production proof load tests shall meet these tests without serious failure or damage. Trailers subjected to preproduction proof load tests shall meet these tests without catastrophic failure although yielding may occur.

3.3.10 Hydraulic lift system endurance. The lift system shall withstand the lift system endurance test in 4.6.12 without deformation, leakage or other damage.

3.3.11 Air transportability. The empty trailer shall be air transportable in accordance with MIL-A-8421. The following exceptions to MIL-A-8421 shall apply:

a. The maximum ramp angle from plane deck shall be 13 degrees.

b. Flight load tests described in MIL-A-8421 shall be replaced by the flight load test in 4.6.13.2 of this specification.

3.3.12 Associated equipment. The trailer shall be capable of lifting all stores, single or multiple, for which designed when the stores are supported by the applicable cradles or bolsters. When applicable, bolsters shall be equipped with their approved lift hook adapter. Complete conformance to this specification and associated drawings will ascertain that the above requirement has been met.

3.3.13 Environment. The trailer shall operate satisfactorily during and after exposure to the following environments:

a. High temperature to 71° Centigrade (C) (160° Fahrenheit (F).

b. Low temperature to -54°C (-65°F).

c. Icing conditions.

d. Sand and dust conditions such as to be encountered in desert locations.

e. Salt spray likely to be encountered through storage or operation in salt sea atmosphere.

3.4 Lubrication. MILGB-21164 grease shall apply for all lubrication unless otherwise specified.

3.5 Identification of product. The trailer, all assemblies, and subassemblies shall be marked for identification in accordance with MIL-STD-130.

3.6 Government-loaned property. The following items of equipment shall be loaned to the contractor, upon his request, as government loaned property for the purpose of testing the trailer:

Item No	Type of Designator	Item Description	Qty
1	MD-3A or equal	Generator, 400 Cycle	1
2	MHU-19/E	Cradle, Bomb Handling	1
3	ADU-317/E	Adapter, Rotary Launcher	1
4	ADU-318/E	Adapter, Pylon	1

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3.7 Workmanship. The trailer, including all parts and accessories, shall be fabricated in a manner that will insure that the criteria of appearance, fit and adherence to specified tolerances shall be observed. These criteria must apply to all accessories and to all external nonfunctioning parts, such as nameplates, as well as the functional parts. Particular attention shall be given to neatness and thoroughness of marking of parts and assemblies, plating, riveting, machine screw assemblage, painting, and welding. All parts shall be free from burrs and sharp edges which might mar associated equipment or injure personnel.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the trailer shall be classified as follows:

a. Preproduction tests

b. Quality conformance tests

4.3 Preproduction tests.

4.3.1 Preproduction test samples. The preproduction test samples shall consist of models representative of the production trailer. They shall be tested at a laboratory designated by the procuring activity or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

4.3.2 Preproduction test report. In the event that preproduction testing is performed at the contractor's plant, the contractor will, after completion of the preproduction tests, prepare a preproduction test report conforming to MIL-STD-831 and furnish three complete copies of the report to the procuring activity.

4.3.3 Preproduction tests. The preproduction tests shall be performed on the first article produced and will consist of all tests described in 4.5 as listed in Table I.

4.4 Quality conformance tests. Quality conformance tests shall consist of the following:

a. Individual tests.

b. Sampling plan and tests.

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TABLE I - TESTS

Test	Requirement Paragraph	Test Paragraph	Preproduction Test	Individual Test	Sampling Test
Examination of Product	3.2	4.6.1	X	X	X
Conformance to Drawings	3.2	4.6.2	X	X	X
Radiographic Inspection	3.3.2	4.6.3	X	X	X
Continuity	3.3.3	4.6.4	X	X	X
Hydraulic Lift System					
Capability Test	3.3.3	4.6.5	X	X	X
Endurance Test	3.3.10	4.6.12	X	X	X
Store Attitude Adjustment Test	3.3.4	4.6.6	X	X	X
Brake System Test	3.3.5	4.6.7	X	X	X
Parking Brakes	3.3.5.1	4.6.7.1	X	X	X
Service Brakes	3.3.5.2	4.6.7.2	X	X	X
Break-Away Brakes	3.3.5.3	4.6.7.3	X	X	X
Steering Test	3.3.6	4.6.8	X	X	X
Castering Test	3.3.7	4.6.9	X	X	X
Road Test	3.3.8	4.6.10	X	X	X
Proof Load Test	3.3.9	4.6.11	X	X	X
Preproduction Proof Load Test	3.3.9	4.6.20	X	X	X
Production Proof Load Test	3.3.9	4.6.13	X	X	X
Air Transportability Test	3.3.11(a)	4.6.13.1	X	X	X
Ramp Test	3.3.11(b)	4.6.13.2	X	X	X
Flight Test	3.3.12	4.6.14	X	X	X
Fit Test	3.3.13	4.6.15	X	X	X
Environmental Tests	3.3.13(a)	4.6.16	X	X	X
High Temperature Test	3.3.13(b)	4.6.17	X	X	X
Low Temperature Test	3.3.13(c)	4.6.18	X	X	X
Icing Test	3.3.13(d)	4.6.19	X	X	X
Sand and Dust Test	3.3.13(e)				
Salt Spray Test					

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4.4.1 Individual tests. Each trailer shall be subjected to the individual tests listed in Table I.

4.4.2 Sampling plan and tests.

4.4.2.1 Sampling plan. The number of sample trailers, indicated below, shall be selected at random from each production lot. Prior to being subjected to sampling test, the sample trailers shall have passed the individual tests. Samples shall be subjected to all sampling tests except the lift endurance test and radiographic inspection. Selection of random samples shall be by the Government's repesentative.

Production Lot	Number of Samples
1 to 49	1
50 to 99	2
100 to 299	3
300 to 500	4

4.4.2.1.1 Samples for lift endurance test. One trailer shall be selected at random from each 300 trailers produced, or from 6 months' production run, whichever occurs first, and subjected to the lift endurance test. Selection of the random samples shall be done by the Government's representative.

4.4.2.1.2 Sampling for radiographic inspection. Prior to assembly to trailers, a random sample of 25 percent of each production lot of all load-bearing steel castings shall be subjected to radiographic inspection in accordance with MIL-I-6865. Selection of the random samples shall be done by the Government's representative. In the event of failure of any casting of this sample, the production lot shall be rejected. Rejected production lots may again be offered for Government acceptance providing:

a. The production lot has been subjected to 100 percent radiographic inspection.

b. All castings not conforming to the requirements of the radiographic inspection in 4.6 of this specification have been removed from the lot.

4.4.2.2 Sampling tests. The sampling tests shall consist of the sampling tests shown in Table I.

4.4.2.3 Rejection and retest. When the sample item fails to satisfy the specified tests, no item still on hand or later produced shall be accepted until the extent and cause of failure is determined and resolved.

4.4.2.4 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of items on hand or produced later shall not be made until it is determined that the items meet all requirements of this specification.

4.4.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.5 Test conditions. Test conditions shall be as described in each specific test in 4.6.

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4.6 Test methods.

4.6.1 Examination of product. Each trailer shall be examined visually to determine compliance with the requirements of this specification for which tests are not specified, including materials and workmanship. Particular attention shall be given to leakage in the hydraulic system, and to routing and protection of hydraulic lines and electrical wiring.

4.6.2 Conformance to drawings. Each trailer shall be inspected to determine conformance to the engineering drawings.

4.6.3 Radiographic inspection. Prior to assembly to the trailer, sample load-bearing steel castings shall be subjected to radiographic inspection in accordance with MIL-I-6865. Criteria for acceptable and unacceptable defects shall be as described in Designation E71, American Society for Testing Materials Radiographic Standards for Steel Casting (X-rays). Failure to meet the requirement of MIL-I-6865 and ASTM Standards shall be cause for rejection of the lot.

4.6.4 Continuity. Using standard instrumentation, determine continuity of all power and lighting circuits. Flood lamp circuits shall be tested in both the retracted and extended conditions. Lack of continuity in any circuit shall be cause for rejection.

4.6.5 Lift system capability test. The following steps shall be taken to test the lifting, holding, and control capabilities of the trailer:

a. Install a 33,500 pound test load on the lifting beams in a manner simulating preparation for aircraft loading.

b. Set the main pump relief valve for a hydraulic pressure of 2700 pounds per square inch (psi) for this test and those of 4.6.6, then reset to 2500 psi.

c. Operate the hydraulic lifting mechanism until the lifting arms are in the fully extended position. Elapsed time required for this operation shall not exceed 180 seconds.

d. Inspect to determine the following:

(1) That the bottoms of the lifting beams are not less than 84 inches from ground level.

(2) That the hydraulic system has functioned in such a manner that the height differential of the lifting beams does not exceed 1-3/8 inches.

(3) That no distortion, deformation, excessive leakage or other damage has occurred.

e. With the lifting beams still in the fully extended position, release the hydraulic lifting pressure. The counter balance valves shall function in such a manner that the lifting cylinders shall hold the lifting beams in position.

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f. Lower the lifting beams alternately in 2 inch increments to the retracted position. During and after this operation, inspect to determine the following:

(1) That full control of both sets of lifting arms, moving together or independently, was maintained throughout operation.

(2) That no distortion, deformation, excessive leakage or other damage occurred.

g. Raise the lifting beams alternately in 2 inch increments to the fully extended position. Inspect as described in 4.6.5f(1) and (2).

h. Lower the lifting beams to the retracted position.

i. Using the auxiliary hand pumps, raise the lifting beams to the fully extended position.

j. Lower the lifting beams until the weight of the loaded trailer is removed from the wheels. Inspect as described in 4.6.5f(1) and (2).

k. Remove the test load and raise the lifting beams to the fully extended position.

l. Lower the lifting be ms to the fully retracted position. Determine that time required for this operation does not exceed 180 seconds.

4.6.6 Store attitude adjustment test. The following steps shall be taken to test the store attitude adjustment features of the trailer:

a. Install a 33,500 pound test load.

b. Raise the lifting beams to fully extended position.

c. Operate the lift mechanism to rotate the load about its longitudinal axis. This operation shall rotate the test load a minimum of 10 degrees clockwise and 10 degrees counter-clockwise about its longitudinal axis. Rotation capability of less than 10 degrees in either direction shall be cause for rejection. Longitudinal axis implies roll axis.

d. Operate the arc compensator mechanism to cause the load to rotate 5 degrees right and left about its own vertical axis (yaw). Failure to rotate the full degrees in either direction shall be cause for rejection.

e. Operate the tilt adjustment mechanism to adjust the nose of the test load to the maximum raised and maximum lowered position. Determine that the tilt adjustment mechanism is capable of adjusting the longitudinal attitude of the load through a range of 5 degrees above and 3 degrees below horizontal. Failure to adjust up to and within these limits shall be cause for rejection.

f. Operate the fore and aft adjusting mechanism to determine that the load can be moved 5 inches forward and 5 inches rearward from center. Failure to adjust up to and within these limits shall be cause for rejection.

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4.6.7 Brake system tests.

4.6.7.1 Parking and emergency brakes. The parking and emergency brakes shall be tested in the following manner:

a. The trailer shall be loaded with a 27,500 pound test load during these tests.

b. Position the loaded trailer on a 13 degree incline, with break-away feature inoperable, and parking and emergency brakes set. The parking and emergency brakes shall prevent movement of the trailer.

c. Gradually release the parking and emergency brakes until the loaded trailer begins to roll down the incline. After two revolutions of the wheels, manually apply the parking and emergency brakes. The trailer shall stop.

d. Connect the loaded trailer to its prime mover and position both on a level dry concrete surface. Set the parking and emergency brakes of the trailer and apply sufficient towing force to pull the trailer. The wheels of the trailer shall skid instead of roll.

e. Failure to satisfy any of these tests shall be cause for rejection.

f. For the purpose of this test, the 13 degree slope may be simulated by applying a 8,500 pound forward force component to the trailer towbar.

NOTE: During the testing described in 4.6.7.1a and c, emergency restraining facilities shall be provided to prevent "run-away" in the event of failure of the brakes.

4.6.7.2 Service brakes. The service brakes of the trailer shall be tested in the following manner:

a. Connect the trailer loaded with a 27,500 pound load, to a prime mover. This connection shall be sufficiently flexible to facilitate observation of independent stopping and slowing of the trailer.

b. Tow the trailer at a speed of 20 miles per hour (mph) on a level, dry, concrete surface. Stop the trailer by applying the service brakes only. The trailer shall stop within 50 feet or less of point where brakes were first applied. Perform the operation five times in the minimum time necessary to attain initial speed each time. At the end of the fifth cycle, inspect the brake disks and actuating mechanism. There shall be no excessive heating or distortion.

c. Tow the trailer down a 10 percent grade. Apply service brakes sufficiently to maintain 5 mph speed for a distance of 1/2 mile. There shall be no damage, distortion of the brakes, or diminished effectiveness of the brakes throughout the test. Grade throughout this test shall be 10 percent.

d. Failure to perform as required in the tests described in 4.6.7.2b and c, or evidence of excessive heating or distortion of the brake disks or actuation mechanism, shall be cause for rejection.

e. For the purpose of this test, the 10 percent grade may be simulated by applying a 6500 pound forward force component to the trailer towbar.

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4.6.7.3 Break-away brakes. The break-away brakes of the trailer shall be tested in the following manner:

a. Connect the trailer, loaded with 27,500 pound test load, to its prime mover with a quick-release provision between the lunette and pintle.

b. While towing the loaded trailer up a minimum grade of 5 percent, at towing speed of 10 miles per hour, trip the quick-release so that all towing connection is removed from the trailer. Break-away brakes shall stop the trailer and prevent it from rolling back.

c. While towing the loaded trailer down a minimum 5 percent grade at towing speed of 10 miles per hour, trip the quick-release so that all towing force is removed from the trailer. The break-away brakes shall operate quickly enough to prevent overrun of the prime mover by the trailer and they shall stop the trailer.

d. Failure to function as specified in 4.6.7.3b and c, above, shall be cause for rejection.

4.6.8 Steering test. The trailer, carrying a 27,500 pound test load shall be towed throughout a sufficient number of right and left turns to determine steering capability specified in 3.3.6. Binding, buckling, or other damage, or a determined steering angle of less than 45 degrees in either direction shall be cause for rejection.

4.6.9 Castering test. The trailer, carrying a 27,500 pound test load, shall be subjected to the following castering test.

a. Lower the lifting arms until the weight of the trailer is removed from the wheels.

b. Manually rotate all front and rear wheels to a position 90 degrees from the loading position.

c. Raise the lifting arms to the traveling position.

d. Move the trailer laterally 10 feet by use of the inching device.

e. Failure of the wheels to rotate smoothly without application of excess force, failure to remain in the castered position during lateral movement of the trailer, shall be cause for rejection.

4.6.10 Road test. The representative trailer shall satisfy the Type II mobility test described in MIL-M-8090, while carrying a 27,500 pound test load.

4.6.11 Preproduction proof load test. The trailer, supported only by the running gear, shall be subjected to a downward vertical static load of 82,500 pounds. This load shall be applied to the lifting arms in the following manner.

a. Once with the lifting arms in the fully retracted position.

b. Once with the lifting arms midway between the retracted and fully raised position.

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c. Once with the lifting arms in the fully raised position.

The tilt and arc compensator mechanisms will be set at their zero positions. Failure of the lift mechanism to support the proof load for 30 seconds shall be cause for rejection.

4.6.12 Lift system endurance test. The trailer loaded with a 27,500 pound test load shall be subjected to a lift system endurance test of 100 cycles performed as follows:

a. Raise the lifting beams to the fully extended position.

b. Lower the lifting beams to the fully retracted position.

c. At the end of step a, above, of each tenth cycle, perform the test described in 4.6.6 except that a 27,500 pound load shall be used.

d. As a continuation of step b, above, of each tenth cycle, lower the lifting beams to the ground and continue to lower until the weight of the trailer is removed from the wheels.

Evidence of distortion or deformation of parts, leakage, or other damage during or after this test shall be cause for rejection.

4.6.13 Air transportability tests.

4.6.13.1 Ramp test. The trailer, carrying a 27,500 pound test load, shall be subjected to the ramp test specified in MIL-M-8090, except that ramp to plane deck angle shall be 12 degrees. It will be necessary to raise the lift arms to perform this test. Failure to meet the requirements of the ramp test shall be cause for rejection.

4.6.13.2 Flight load test. The trailer, without payload, shall be secured in a manner simulating tiedown in a cargo aircraft. While in the air transport configuration, the trailer shall be subjected to the following loads applied in the manner indicated. Direction of application shall be in reference to the aircraft:

a. 3.00 g forward, 1.5 g upward - applied simultaneously.

b. 3.00 g forward, 1.5 g laterally - applied simultaneously.

c. 4.00 g forward.

d. 1.50 g aft.

e. 1.50 g laterally, left to right, and right to left.

f. 3.68 g upward.

These loads shall be applied once with the front of the trailer toward the front of the aircraft, and once with the rear of the trailer toward the front of the aircraft.

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g. Operate the lift and brake mechanisms.

Failure to operate satisfactorily, or failure or permanent deformation of any part shall be cause for rejection.

4.6.14 Fit test. The trailer shall be fit tested by installing a sample of each type of fixture or adapter to be used with, or loaded on, the trailer in the same manner as in actual service. Binding or misplacement attributable to the location, size or finish of attachment or rest points of the trailer shall be cause for rejection.

4.6.15 High temperature test. During this test, temperature tolerances shall be in accordance with MIL-STD-810. The trailer shall be lubricated with grease conforming to MIL-G-21164 and the electric motor gear box shall be lubricated with lubricating oil conforming to MIL-L-6086, Grade M, prior to being subjected to the following tests:

a. The trailer, loaded with a 27,500 pound test load, shall be placed in a high temperature chamber and the temperature raised to 71°C (160°F) with a relative humidity of not more than 5 percent. The trailer shall be maintained at a temperature of 71°C (160°F) for a period of 48 hours. While at this temperature, the lift, attitude adjustment, and brake mechanisms shall be operated a sufficient number of times to determine that it will operate satisfactorily for a period of 10 minutes at the temperature of 71°C (160°F). The trailer will be operated at least four times.

b. Lower the temperature of the trailer to 54°C (130°F). While the trailer is maintained at this temperature, operate the lift, attitude adjustment, and brake mechanisms continuously for a period of two hours at the temperature of 54°C (130°F).

c. Lower the temperature to 49°C (120°F). While the trailer is maintained at this temperature, operate the lift, attitude adjustment, and brake mechanisms continuously for a period of two hours at the temperature of 49°C (120°F).

Failure to operate satisfactorily and at specified speeds during this test shall be cause for rejection.

4.6.16 Low temperature test. The trailer shall be lubricated with grease conforming to MIL-G-21164 and the electric motor gear box lubricated with lubricating oil conforming to MIL-L-6086, Grade L, prior to being subjected to the following:

a. The trailer, loaded with a 27,500 pound test load, shall be subjected to the low temperature test described in Procedure I of MIL-STD-810. Reasonably reduced rate of lift will be acceptable when undiluted hydraulic fluid is used. Operation check shall include the attitude adjustment and brake mechanisms.

Failure to operate satisfactorily a total of four times during and after this test shall be cause for rejection.

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4.6.17 Icing test. During this test, temperature and relative humidity tolerances shall be in accordance with MIL-STD-810. The trailer, lubricated as specified in 4.6.16, shall carry a 27,500 pound test load while being tested in the following manner:

a. Place the trailer in a chamber having a temperature of 49°C (120°F) and relative humidity of 95 percent. These conditions shall be maintained for a period of 48 hours.

b. Move the trailer into a rain test chamber. A temperature of 20°C (68°F) to 30°C (86°F) shall be maintained in the rain test chamber throughout this portion of the test. A simulated rainfall of 4 ±1 inch per hour shall be produced by means of a water spray nozzle of such design that the water is emitted in the form of small droplets rather than a fine mist. The temperature of the water shall be maintained between 11°C (51.8°F) and 20°C (68°F). The rainfall shall be dispersed uniformly over the test area. The trailer shall be subjected to these conditions for a period of 30 minutes.

c. Immediately after subjection to the conditions in step b, above, the trailer shall be placed in a chamber which has been cooled to a temperature of -18°C (0°F). The trailer shall remain in this environment for a period of 48 hours.

d. Immediately after the exposure in step c, above, and while at a temperature of -18°C (0°F), all mechanisms shall be operated a total of four times.

Failure to operate satisfactorily while at -18°C (0°F) shall be cause for rejection.

4.6.18 Sand and dust. The trailer shall be subjected to the sand and dust test described in MIL-STD-810. Immediately after being subjected to this test, all mechanisms of the trailer, loaded with a 27,500 pound test load, shall be operated a total of four times. Failure to operate satisfactorily after this test shall be cause for rejection.

4.6.19 Salt spray test. The trailer shall be subjected to the salt spray test described in MIL-STD-810. At the conclusion of this test, all mechanisms of the trailer, loaded with a 27,500 pound test load, shall be operated a total of four times. Failure to operate satisfactorily shall be cause for rejection.

4.6.20 Production proof load test. The trailer, supported only by the running gear, shall be subjected to a downward vertical static load of 55,000 pounds. This load will be applied in the same manner as that of 4.6.11 of this specification. Failure of the lift mechanism to support the load for 30 seconds shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Preservation, packaging and packing for levels A, B, and C shall be in accordance with MIL-STD-162 except that the crate for overseas shipment shall be in accordance with MIL-C-3774.

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5.2 Marking. Accessory and interior packages and exterior containers shall be marked in accordance with MIL-STD-129. In other than overseas shipment, the nameplate of the trailer shall be sufficient marking for the trailer proper. The nomenclature shall be as follows:

TRAILER, BOMB MHU-123/M

6. NOTES

6.1 Intended use. The trailer covered by this specification is intended to be used for transporting weapons from storage areas to a loading position under a bombing aircraft and raising the weapon to loading position in or on the aircraft.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Conditions under which Government Loaned Property will be made available to the contractor.

c. Whether preproduction tests will be conducted by the procuring activity or by the contractor under the supervision of the procuring activity.

d. Required levels of preservation, packaging and packing.

6.2.1 Preproduction test samples. It is expected that the contract or purchase order will specify a minimum of 2 trailers representative of the production trailer will be submitted to the procuring activity for test and evaluation. The invitation for bids, and the contract or purchase order should specify the point of inspection for these tests.

6.2.1.1 Destructive test. Two preproduction test samples are enumerated above due to the fact that the flight load test could be destructive, or at least could render the sample item unserviceable for further testing.

6.3 Production delivery. No production items should be delivered prior to approval of the preproduction samples.

6.4 Definitions.

a. Preproduction tests. Preproduction tests are those tests made on samples which have been submitted by a contractor on a production contract or purchase order to prove that the equipment meets the requirements of this specification.

b. Acceptance tests. Acceptance tests are those tests accomplished on equipment submitted for acceptance under a contract to determine that the units are equivalent in performance and construction to the approved preproduction samples.

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c. Production lot. This definition applies to steel castings offered for sample inspection under the requirements of 4.3.2.1.2 of this specification. A production lot shall consist of castings made from the same melt or the same pouring of a ladle containing a number of smaller melts, heat treated in the same furnace charge, and offered for inspection at one time.

d. Excessive leakage. No leakage is allowed at any joints regardless of the ambient temperature. The cylinder leakage rate at 9-65°F shall not exceed 1/4 pint of liquid per 5 cycles of operation where motion through external packing exists. All joints of the cylinder shall have no leakage regardless of the temperature.

Custodian:
Air Force - 79

Review Activity:
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Preparing Activity:
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OMB Approval
No. 22-R255

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